

VERNADSKY BASE, Antarctica

WMO: 890630

Lat: 65.246S

Lon: 64.258W

Elev: 36

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-2.7	2.0	-9.8	3.2	-1.4	-4.2	4.4	3.7	43.5	32.7	37.7	28.5	7.4	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.4	41.1	36.7	39.6	35.7	38.4	35.0	37.9	39.9	36.7	38.6	35.8	37.5	20.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
36.3	31.5	37.8	35.1	30.0	36.7	34.2	29.0	36.0	14.4	40.3	13.9	38.7	13.5	37.7	42.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.7	32.0	28.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	34.0	34.2	32.8	29.1	27.4	25.3	22.2	19.7	21.7	25.5	28.2	31.4		
	DBStd	7.29	2.31	2.61	2.84	3.99	4.60	4.87	6.69	8.54	9.63	6.96	5.12	2.40		
	HDD50	8182	497	443	532	628	700	742	862	940	850	760	653	575		
	HDD65	13657	962	863	997	1078	1165	1192	1327	1405	1300	1225	1103	1040		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	9.7	5.9	8.9	10.9	10.0	10.1	11.3	10.8	10.0	12.4	10.5	9.4	6.5		
	PrecAvg	30.0	2.4	2.5	3.0	2.5	2.4	2.0	2.2	2.3	2.7	3.1	2.8	2.1		
	PrecMax	39.2	4.4	4.0	4.4	4.5	4.5	3.5	4.0	3.5	3.9	5.1	4.5	4.7		
	PrecMin	21.5	0.9	1.5	1.6	1.1	1.5	1.0	0.9	0.8	1.5	1.7	1.4	0.7		
	PrecStd	4.9	0.9	0.6	0.6	0.7	0.7	0.6	0.7	0.7	0.6	0.8	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	44.1	42.2	40.2	41.7	37.2	36.5	38.2	37.6	40.0	39.3	40.0		
		MCWB	37.9	37.9	37.5	36.8	36.0	32.9	31.3	33.6	33.9	36.0	36.1	34.6		
	2%	DB	40.4	40.5	39.8	38.2	38.2	35.0	34.4	35.3	35.5	37.3	36.4	37.8		
		MCWB	36.4	36.8	35.6	34.9	33.8	31.6	31.0	32.4	32.5	34.4	34.0	34.0		
	5%	DB	38.7	38.9	38.1	36.2	35.4	33.1	32.7	32.7	34.0	35.6	35.1	36.3		
		MCWB	35.5	35.7	34.9	33.8	32.4	30.6	30.3	30.5	31.8	33.1	33.0	33.4		
	10%	DB	37.3	37.8	36.7	34.5	33.4	31.5	31.3	30.6	32.7	34.0	33.8	35.0		
		MCWB	34.5	35.2	34.2	32.3	31.5	29.5	29.4	28.5	31.1	32.0	31.9	32.7		
	0.4%	WB	39.6	39.5	38.3	38.0	37.1	33.2	32.9	34.2	35.0	37.5	36.5	36.7		
		MCDB	41.6	41.6	41.0	39.4	42.2	36.2	34.3	37.1	36.2	39.1	37.8	38.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.4	37.8	36.6	35.5	34.2	32.1	31.6	32.4	33.4	34.9	34.7	35.2		
		MCDB	39.2	39.7	38.6	37.3	37.0	34.4	33.3	34.3	34.5	36.4	36.0	36.5		
	5%	WB	36.0	36.6	35.5	34.2	32.7	31.1	30.5	30.9	32.3	33.5	33.5	34.1		
		MCDB	37.7	38.4	37.4	36.0	34.9	32.9	32.3	32.6	33.6	35.2	34.6	35.5		
	10%	WB	35.2	35.7	34.5	32.7	31.7	30.0	29.4	28.6	31.4	32.2	32.5	33.2		
		MCDB	36.8	37.3	36.1	34.3	33.2	31.5	31.1	29.9	32.7	33.6	33.5	34.6		
	5% DB	MDBR	4.6	4.4	4.4	4.2	4.8	5.6	6.7	8.1	9.7	7.7	6.8	5.2		
		MCDBR	6.9	6.2	7.0	6.2	8.3	6.6	7.8	8.4	8.6	7.5	7.0	7.3		
		MCWBR	4.7	4.0	4.8	5.0	5.0	4.8	6.7	6.3	7.6	5.7	5.6	5.1		
		MCWBR	6.1	5.8	6.2	5.7	7.5	6.3	7.3	8.7	9.2	7.2	6.6	6.3		
	5% WB	MCWBR	4.5	4.2	4.7	5.1	5.2	5.0	6.8	6.9	8.8	6.0	5.6	5.1		
		taub	0.284	0.281	0.266	0.237	N/A	N/A	N/A	0.233	0.258	0.276	0.279	0.284		
		taud	2.383	2.402	2.412	2.311	N/A	N/A	N/A	2.161	2.266	2.258	2.303	2.342		
		Ebn at Noon	311	294	268	211	N/A	N/A	N/A	201	263	292	310	315		
		Edh at Noon	33	29	22	14	N/A	N/A	N/A	16	25	33	36	36		
(44) All-Sky Solar Radiation	RadAvg	1926	1293	681	241	49	8	30	165	563	1249	1912	2226			
	RadStd	114	80	49	19	2	0	2	13	51	73	121	181			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page